From S13 to the current VSL model

One traffic chain, with the changes made explicit

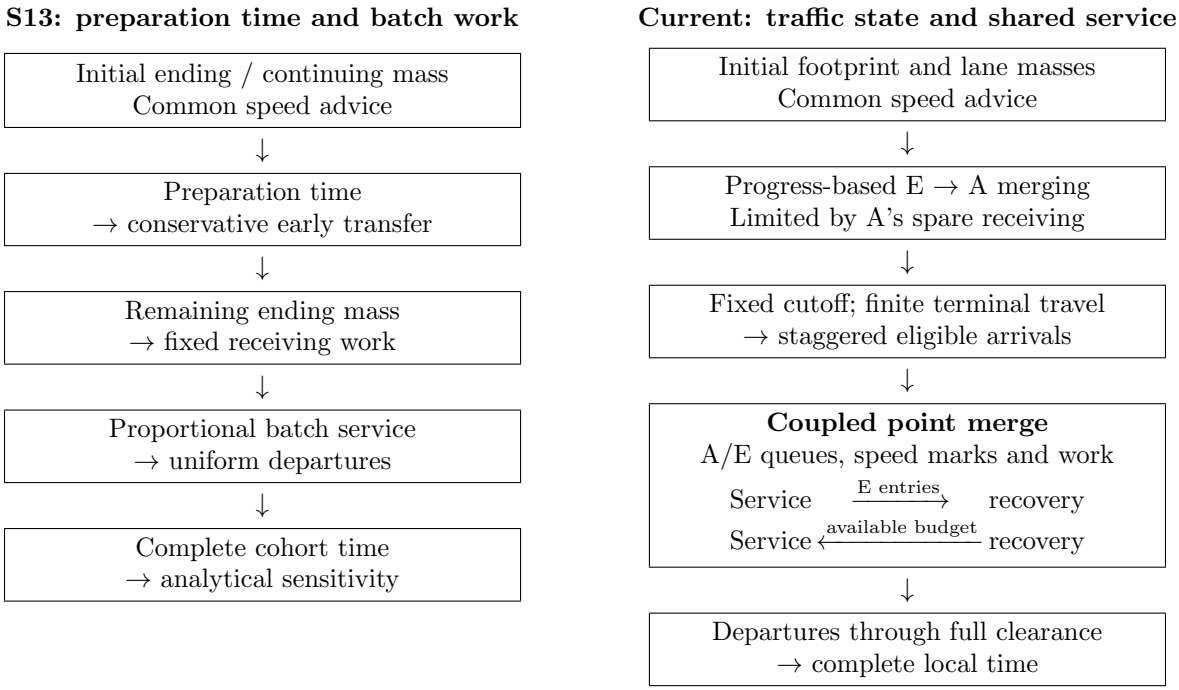
Feedback 8 recap — 16 October 2026

1. The backbone that remains

The task is still a permanent three-to-two lane drop with one finite traffic cohort and one constant speed advice shared by all lanes. The research asks whether information about unfinished mandatory merging helps predict a useful speed decision. It does not assume that a full lane representation, vehicle identity or discrete order is necessary.

S13's central idea was simple: slower travel gives traffic more time to prepare for the lane drop; better preparation can reduce the work required at the final merge. The total-time objective charges the slower travel as well as subsequent service.

The current stack keeps the control question while changing the traffic mechanism: progress and spare receiving govern early transfer; timed arrivals feed a coupled choke; admitted E can restrict later service. The two backbones can therefore be read side by side:



E is the ending lane; A is its adjacent continuing lane. In the current choke, E entries update recovery, which changes the budget for later service. The arrows inside that box form a causal feedback loop.

Reading the status correctly. This is a functionally complete *conditional mathematical model*. The selected formulas and their forward arguments are assembled here; this recap changes none of them. Physical parameter admission, the amended information/action and gradient protocol, and the final SUMO correspondence remain unfinished. Step 3 stays open at MP6-Q1; Step 4 is unopened. No implementation or experiment is reported.

2. What changed: setting, motion and early merging

The component numbers preserve the original nine-component audit. Several equations belong to the same component; they are not additional project stages.

Original component	Old → current; the core question it now answers
1. Geometry, population and preparation	**Modified.** S13 grouped both continuing lanes and placed the cohort at one preparation entrance. The current model distinguishes ending E, adjacent A and far continuing B, with a uniform occupied footprint and an initially empty terminal. *Which lane receives the merge, and where is the traffic when advice begins?* Initial A/B symmetry remains; their later states need not be equal.
2. Advice and response	**Retained with an explicit boundary.** One bounded, constant common advice remains. It sets the ideal free-motion speed in the admitted branch; queued boundary marks can be lower. *Through which physical state can advice influence service?* The final work law has fixed coefficients and no direct advice multiplier. Instantaneous native compliance is unverified.
3. Longitudinal timing	**Replaced.** One shared traversal time followed by batch service becomes position-dependent free travel, a fixed early-merge cutoff, and finite terminal travel/storage represented by boundary histories. *When does each part of the cohort become available at the choke, and can its queue reach preparation?* No new spatial grid is introduced.
4. Mandatory preparation transfer	**Replaced.** A constant completion fraction per second becomes a homogeneous offer limited by donor traffic and A's spare receiving. *How much early merging can actually succeed?* A fixed physical opportunity length replaces an unqualified elapsed-time completion law.
5. Preparation receiving	**Made explicit.** S13 assumed it did not bind. A's own traffic now claims receiving first; E uses the remaining work opportunity. Conservative terminal storage bounds protect this homogeneous upstream calculation. *Can A accommodate the attempted transfers without an omitted upstream queue?*

Components 4 and 5 form one joined preparation component in the present equations. They are listed separately to show which original assumptions changed. Distinguishing A from B does not create new initial information requirements under the retained initial symmetry.

3. What changed: final merging, accounting and evidence

Original component	Old → current; the core question it now answers
6. Choke service and departures	**Replaced and expanded.** Fixed receiving work and proportional batch service become a causal A/E node. It uses eligible queues, free/queued speed marks, current maneuver work, a priority/cooperation rule and recent E recovery history. *Who can cross now, and how does an admitted slow merge affect later crossing?* A can now wait as well as E. B remains independent in the admitted domain.
7. Complete-time objective	**Meaning retained; calculation replaced.** Full affected-cohort time remains the objective, but the old batch formula is replaced by actual departure or queue-area integration. *Does every unit's travel and waiting count exactly once?* Completion is at the choke; downstream physical influence may remain after that unit's own clock stops.
8. Information and initialization	**Retained and rechecked for the forward law.** Aggregate history, that history plus initial E mass, and equivalent explicit lane state remain the comparison ladder. *What extra initial measurement is sufficient?* Known total mass plus initial E mass still reconstructs this member. Future queues and recovery history are predicted internally.
9. Sensitivity and decision evidence	**Standard retained; result must be requalified.** The physical common advice is still the differentiated variable. *Does the complete objective's response support a useful decision?* S13's analytical action split remains valid for S13, but does not automatically transfer to these changed laws. The current stack may have nonsmooth activation events.

The genuinely new dynamic content is concentrated in component 6: A's waiting balance, the speed/work coupling, and the retained recovery account. Finite terminal histories also add timing and storage information to component 3. None of these is a microscopic gap or vehicle-identity model.

SUMO is an external evidence layer, not a component needed to calculate the primary forward trajectory. The agreed ladder is primary-model verification/optimization, restricted SUMO S-P, then natural late-merging SUMO S-L transfer. Its amended protocol remains to be completed.

4. The new preparation law: opportunity plus spare receiving

Use E for the ending lane, A for its adjacent continuing lane and B for the far lane. Let $e(t), a(t), b_0$ be densities in the homogeneous upstream body. Put $r = e_0 + a_0$, with $a_0 = b_0$, and let $h > 0$ be a fixed physical opportunity length. Define

$$K = wk_j, \qquad k_c(v) = \frac{K}{w+v}, \qquad C(v) = vk_c(v).$$

Here w is a backward-wave speed and k_j jam density. These define the selected triangular traffic relation; they are not fitted in this recap.

The core rule. A uses its own receiving first. E's accepted opportunity-equivalent rate f is the smaller of its offer and the spare receiving available after that claim:

$$S(v,a) = \min\{C(v), w(k_j - a)\}, \qquad f = \min\left\{ve, \frac{S(v,a) - va}{\gamma}\right\}, \qquad \dot{e} = -f/h, \quad \dot{a} = f/h.$$

$\gamma \geq 1$ is early-merge receiving work. Equal mass leaves E and enters A. f is not an extra longitudinal boundary flow: the actual lateral source per unit road length is f/h.

In the admitted free upstream branch, $S = C$ and $e + a = r$. Expressed per distance traveled, $z = vt$, the same rule is

$$\frac{de}{dz} = -\frac{1}{h}\min\left\{e, \frac{\Delta(v) + e}{\gamma}\right\}, \qquad \Delta(v) = k_c(v) - r.$$

This exposes an important change from S13. If donor availability alone limits transfer, the residual is exponential in *distance*, so slower motion alone gives no extra preparation over the same distance. A speed effect survives through the declared receiving headroom when that limit matters. Longer residence by itself is no longer assumed to guarantee better preparation.

How this becomes arrivals. The preparation ends at L_p, before a terminal of length b, with $L = L_p + b$. Initial positions occupy $[0, \ell]$. Over the handoff interval $((L_p - \ell)/v, L_p/v)$,

$$\dot{F}_E = ve(t), \quad \dot{F}_A = v(r - e(t)), \quad \dot{F}_B = vb_0; \qquad P_j(t) = F_j(t - b/v).$$

F_j counts terminal entries and P_j counts traffic that has had time to reach the choke. Early merging stops at L_p; remaining E must merge at the choke. This is the selected opportunity restriction, not a claim that native drivers universally avoid late lateral opportunities.

5. The new point service: traffic states determine current work

The terminal is a finite physical approach, represented by cumulative histories rather than interior cells. For each lane,

$$N_j = F_j - Y_j, \qquad q_j = P_j - Y_j, \qquad g_j = \dot{Y}_j.$$

N_j is traffic physically in the terminal. q_j is traffic eligible to cross but still unserved. They are different quantities. Queues can have physical extent; a common service point does not make them vertical stacks or imply a stationary queue tail.

Speed marks. Free traffic uses $u_j = v$. A queued approach uses the equilibrium boundary mark

$$u_j = U(g_j), \qquad U(g) = \frac{wg}{K - g}.$$

This is a representative crossing-speed assumption, not a tracked vehicle speed or a whole-lane average. Its physical accuracy remains to be tested.

Immediate work. With the available shared account $\mathcal{B}(t)$,

$$c_A = 1, \qquad c_E = c_0 + \chi g_A[u_A - u_E]_+, \qquad c_0 = 1 + \delta \geq 1, \qquad g_A + c_E g_E \leq \mathcal{B}.$$

c_E is work per E mass. It includes a baseline maneuver burden and extra work when passing A is faster than E. The coefficient χ is fixed: advice influences this expression through traffic state, not through a moving normalizer. There is no mismatch burden without passing A traffic.

Who gets the service? If A is free and its offer fits, it receives its offer first. E takes feasible residual work. If A queues, cooperation gives each backlogged stream equal work; unused work goes to a stream that can use it. Equal work need not mean equal vehicle flow. For queued E, define the unique service root by

$$H(g; a, u) = g\{c_0 + \chi a[u - U(g)]_+\} = x.$$

Free A uses $(a, u, x) = (\lambda_A, v, \mathcal{B} - \lambda_A)$. Both queued use $(a, u, x) = (\mathcal{B}/2, U(\mathcal{B}/2), \mathcal{B}/2)$. Here $\lambda_j = \dot{P}_j$.

For example, free fast A beside queued slow E produces a large mismatch burden and scarce E service. If A also queues, both marks are lower and cooperation becomes active. A retains an advantage from E's maneuver cost. Cooperation remains applicable while A queues even if E clears first; it is not a literal two-positive-queues toggle.

An empty approach stays free while its offer fits. A strict service shortfall starts waiting; true clearance retests free admission. Equality alone does not create a queue, and no numerical queue threshold is added.

6. The new recovery account: one reading, sufficient history

Current entry and its later influence are two parts of the selected point model. After a slow E entry, following traffic can remain restricted even though that E traffic has crossed the completion line. The model retains this influence in one shared account:

$$\mathcal{B}(t) = [R - p_{\mathrm{rec}}(t)]_+.$$

R is undisturbed shared service. The current model derives a strictly positive lower budget bound in its conditional domain; it does not insert an artificial service floor.

A single entry's remaining influence. With physical acceleration parameter $a_r > 0$, an admitted speed mark u generates

$$V(\theta; u) = \sqrt{(w+u)^2 + 2wa_r\theta} - w, \qquad D(\theta; u) = [R - c(V(\theta; u))]_+, \quad c(s) = \frac{Ks}{w+s}.$$

The profile decreases as it ages and becomes zero after a finite time. V is a transformed boundary quantity used to calculate passing capacity; it is *not* the vehicle's speed trajectory. u, by contrast, is the model's representative insertion-speed mark.

How entries combine. Only E admissions create new profiles. Let $T(\xi)$ be the time E mass rank ξ crossed. With fixed physical normalization $m = 1$ vehicle-equivalent,

$$p_{\mathrm{rec}}(t) = \frac{1}{m} \int_{[Y_E(t)-m]_+}^{Y_E(t)} D(t - T(\xi); u_E(T(\xi)))\, d\xi.$$

New E mass adds its profile and retires equal oldest weight once the recent-mass window fills. Retained profiles continue aging. With no new E, recovery continues. There is no per-step reset, queue-clearance reset, or deletion of traffic when an old profile retires.

This makes the service reading a singleton, but its future generally requires recent age/mark information. Two equal readings may recover differently. That memory is generated by the model and is not extra future information supplied to the controller.

The approximation can retire an old strong restriction when newer fast E passes. A does not independently deposit a second restriction merely because it is slowed by the existing one. Both are selected simplifications whose physical credibility remains open. They are not exact microscopic leader or gap dynamics.

7. What is counted, and what information initializes it?

The objective remains complete cohort time. Every initial unit is counted until it crosses the choke. With total mass M,

$$J(v) = \int_0^\infty \left(M - \sum_{j=E,A,B} Y_j(t) \right) dt = \frac{M(L - \ell/2)}{v} + \int_0^\infty (q_A + q_E)\, dt.$$

The first term is travel from the uniform initial footprint. The second counts all excess terminal waiting, including A's new delay. B stays unqueued under its adequate independent service condition. Dividing by M gives mean time and preserves action rankings.

The old S13 formula used half the batch clearance duration because its departures were uniform over that duration. The current model cannot replace queue area by half of final clearance: arrivals, priority and recovery now determine the departure curve.

Completion and physical influence are different. Once an E unit crosses, its own local time is finished. Its retained recovery profile can still hold back unfinished A/E traffic. The waiting of those followers is counted. Post-line travel time of the completed unit is outside the chosen local objective. Recovery memory is not traffic mass.

One extra initial scalar is still sufficient. Let E_0 be initial ending mass. Known M, ℓ and initial A/B symmetry give

$$e_0 = E_0/\ell, \qquad a_0 = b_0 = (M - E_0)/(2\ell).$$

Known empty terminal and recovery histories complete initialization. The model then generates its own preparation, arrivals, queues and recovery. Aggregate-plus-E_0 and explicit lane state therefore make the same forward prediction for this member. Full lane state is not established as a necessary remedy; the current meter reading alone is not sufficient evolving state.

Why the speed tradeoff is still an open result. Speed affects free travel, preparation headroom, arrival times and potentially free-A entry mismatch. Recovery couples later service to earlier E admissions. These paths can oppose one another. A slower optimum or useful gradient does not follow merely because all paths are present. The earlier S13 action-split certificate belongs to its original equations.

8. What has been established, and at which level?

Question	Current answer
Is there a complete forward law?	Yes conditionally: inputs and predicted history determine preparation, entry, service, queues, complete departures and TST throughout the original advice interval under the retained bounds.
Can merging delay A?	Yes in a constructed complete analytical family, with positive recovery and mismatch, ordinary underload and active renewal. This is not a calibrated parameter range or a preferred-action result.
Can storage protect homogeneous preparation?	Yes under the sufficient finite-storage and entry-capacity conditions. The model does not thereby solve arbitrary upstream spillback.
Is extra initial information compact?	Yes. Total mass plus initial E mass and symmetry reconstruct this member. A richer representation has no promised advantage at the same information.
Is the joined physical regime admitted?	No. Crossing-speed interpretation, preparation scales, current work and recovery need joint evidence with one parameter set. A good TST fit alone cannot qualify their separation.
Are useful actions and gradients proved for this stack?	No. Their amended review and independent evaluation remain. Forward uniqueness does not imply smooth sensitivity or an interior optimum.

The forward proof exploits the monotone H1 arrival pulse and selected recovery/priority rules. E can have at most one waiting episode in that domain; repeated A events have a positive separation. A finite cohort *alone* does not imply that E cannot requeue. Bursty later E arrivals can requeue in a wider setting or native SUMO, which must be interpreted as possible model discrepancy rather than automatically rejected.

A useful negative control is already established: remove the baseline extra maneuver work ($\delta = 0$), and ordinary underload keeps the primary cohort free. Mismatch and recovery never activate, so fastest advice is common. Positive information or optimizer results are not required.

9. Literature models that are serious rivals

The review has identified the following comparison candidates. They are established model families with relevant mechanisms, not yet a frozen set of benchmarks for this exact observation, action and objective contract.

Aggregate history and capacity drop — Wei, Han and Wang (2025). A first-order link model retains cumulative counts and entry-speed history, with VSL effects and traffic-state-dependent discharge loss. This is a close aggregate rival: useful memory and endogenous service need not require lane decomposition. It is stronger than a density-only, fixed-capacity peer.

Compact lane-drop capacity control — Jin and Jin. An aggregate lane-drop model represents a discontinuous reduction in discharge and studies VSL control. It offers a simpler explanation of a speed benefit through avoiding capacity loss. Its idealized limits also help identify conditions where VSL has no benefit. Its capacity-drop branch needs its own treatment before gradient comparisons.

Dynamic aggregate speed and predictive control — Hegyi, De Schutter and Hellendoorn (2005). METANET predicts segment density and mean-speed dynamics under speed limits; a predictive controller uses these to manage congestion. It provides an aggregate control rival with evolving speed rather than an instantaneous free-motion response. Its repeated decisions differ from our constant advice.

Lane-aware receiving and priority — Carey, Balijepalli and Watling. A multi-lane CTM couples longitudinal and lateral transfers through finite target receiving and extra lane-changing headway, with alternative priority rules. It is both a source lineage and a fuller continuous lane-aware rival. The published model does not itself supply our common-VSL comparison.

Density-driven mandatory merging — Nagalur Subraveti, Knoop and van Arem. Lane densities, lane-end incentives and receiving constraints determine realized transfers. This is a spatial continuous alternative to homogeneous preparation. Its inspected validation concerns lane-density prediction; common-advice response and gradient usefulness require separate qualification.

Analytical speed-control gradients — Delle Monache, Piccoli and Rossi. An LWR speed-control formulation derives cost variations for outflow tracking. It is a gradient-method reference, rather than a matched lane-drop task: its control and objective differ. VSL gradients themselves are established.

Comparison principle. Published models test alternative traffic explanations. Aggregate-plus-initial-E tests information sufficiency, while bounded scalar search and finite differences test optimization methods. Keep these roles distinct and match the admitted inputs, action and full time account before attributing a performance difference to information.

10. How to use the two reports

This recap answers the framework and old-to-new questions. The companion report supplies the equations, units, admissible domain and derivations in one notation, including conservation, root uniqueness, recovery, event continuation, full clearance, the complete analytical A-delay construction and the physical-identification limits.

Companion: consolidated mathematical report

The mathematical report distinguishes the historical S13 certificate from the selected current law. Historical rejected alternatives are not silently combined into a primary model. It records $\mathcal{B}$ for the shared budget and p_{rec} for its restriction; older reports often write B and p. This is a notation clarification only, avoiding confusion with far lane B.

Source basis and adaptation. The component chain draws on the already reviewed lane-conservation/mandatory-transfer literature, receiving- work and allocation literature, boundary-count traffic propagation and point-insertion recovery. H1's homogeneous law, the selected speed/work closure, priority guards and recent-mass renewal are declared adaptations. No cited paper supplies this complete joined stack. The detailed source and proof map is retained separately:

Archived source, equation and decision map

For an eventual control study. Compare full aggregate history, that history plus initial ending mass, and explicit lane state at the same information. Keep capable aggregate, history and capacity-drop models, analytical and finite-difference gradients, and bounded scalar search as serious rivals. Correct differentiation, improved internal cost and external policy usefulness remain different claims.

Immediate disposition. Feedback 8 is a consolidation request, not authorization to resume the unfinished protocol reviews. This delivery finishes the recap and full mathematical reporting. The active scientific checkpoint remains MP6-Q1; applied MP6 and Step 3 stay open. After review, the next allowed review work is the information/action/full-objective gradient protocol and then the updated model $\rightarrow$ SUMO S-P $\rightarrow$ SUMO S-L contract. Step 4, simulator development, calibration and traffic experiments remain unopened. S13, the urban task and the ramp line retain their closure.